BIT1123/BISE2093/DIT1113

BIT1123 OBJECT ORIENTED PROGRAMMING

ASSIGNMENT 1 – INDIVIDUAL

(20%)

ALL SECTIONS

FACULTY OF INFORMATION TECHNOLOGY

CITY UNIVERSITY MALAYSIA

CYBERJAYA CAMPUS

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1. Introduction

BIT1123 Object Oriented Programming gave me an introduction to concepts and practical application of Object Oriented Programming (OOP) in the language Java. I learned about the development of java programs throughout the tutorials making use of classes and objects, constructors, methods and relationships between classes. The tutorial work should be gathered in a single repository on GitHub together with documentation and a self-reflective report and be part of the assignment.

The tutorials for Java that I did while I took this course are in my GitHub repository aymanhussein5677/aymanhussein202505010362 and show the progress I have made while taking this course. The exercises guided me from basic java concepts to advanced java concepts that were based on OOP.

The tutorials also gave me insight on the fact that programming is not only about writing code. Organizing files, documenting projects and utilizing version control and reflecting on and analyzing problems that arose during software development are also part of good software development.

Tutorial 1

In Tutorial 1, I was introduced to the basics of the structure of a Java class and how classes can be used to instigate objects. With the Student example, I got an understanding of attributes, constructor and methods.

This tutorial was crucial as it was my introduction to simple concept of OOP. I learned that a class can be a blue print and an object can be created from that class. I also learnt that information can be stored within an object and methods used to carry out actions.

Tutorial 2

In Tutorial 2, I built upon the knowledge I gained in Tutorial 1 by completing the Student.java and main.java exercises.

The tutorial has enabled me to get more familiar with making objects and testing the behavior of a class with a main program. I was told that the program can be easier to understand and manage if a class is separated from the main program.

I also learned about how to write java code and what to do when the java program didn't run the way I wanted it to.

Tutorial 3

Examples with Person, Student and Lecturer classes were introduced in Tutorial 3.

This tutorial made me see how objects of different classes can be used to represent different types of objects in a program. I realised that Object Oriented Programming has advantages for modelling real world entities and their interactions.

In working with these classes, I also learned that the same types of objects can have similar properties, but also unique information and behaviour.

Tutorial 4

It is not possible to verify Tutorial 4 since a clearly identifiable tutorial4 folder was not found in the repository that was examined for this report.

So, if I make the source code and support files for Tutorial 4 separately, then I should add them in. This is relevant since the tutorial work from the appropriate weeks must be part of the repository for this assignment.

Tutorial 5

Tutorial 5 included a Student exercise, a main program and documentation-related material.

This tutorial provided me a way of continuing my practice of the java class-based programming. I also learned about the value of documentation. It's okay to write code but explaining and organizing code is also important to have for a software project.

The tutorial has helped me to gain confidence in writing and testing java classes.

Tutorial 6

There were employee classes and a main program in Tutorial 6.

This tutorial cleared up some of my understanding of relationships between classes and provided me with more practice in concepts related to inheritance. I learned that a general class can have common information and a more specific class can be a specialized type.

This helped me to understand the usefulness of "Inheritance" in Object Oriented Programming.

Tutorial 7

Tutorial 7 included classes such as Applience, Microwave and WashingMachine.

With this exercise, I gained an understanding of how a general class can be represented, and more specific types with child classes.

It also helped me to understand how classes may be structured so that similar attributes do not need to be duplicated.

Tutorials 8–9

For this part of the tutorials, there is practical work in the repository: main.java and task.txt.

The tutorials provided me the necessary avenues to get more hands-on experience on implementing the concepts of programming in java. Solution of programming problems involves careful reading of problem, breaking into smaller steps, testing the program after making changes.

The exercises also made me feel more familiar with the java syntax and program structure.

Tutorial 10

Tutorial 10 provided with the book contained two java documents: QuizBattleGUI.java and Questions.java, these were the start of a java GUI Quiz application.

This was also an important step since it demonstrated to me how the concepts of OOP that I have learnt can be applied to create a more interactive application. Although I usually only work with simple console programs, I was able to visualize how one could use Java to develop an application that interacts with users.

This tutorial also helped me get an idea of how multiple components and classes need to be used together to operate the larger program.

3. Challenges Encountered

I had a few difficulty with this: Understanding the difference between basic programming and Object Oriented Programming.

Initially, I had a better time thinking in individual statements and instructions. But in OOP, I had to consider the concepts of classes, objects, attributes, methods and relationships between classes. Practice was needed to see how these various parts function together.

Other troubles were with constructors and methods. I needed to grasp what the use of constructors when creating an object is and how a method enables an object to take on a certain action.

Another difficult idea to come to terms with was the idea of inheritance. The parent-child relationship between the parent class and child class had to be understood and, of course, the question had to be resolved about what information is part of the parent class, and what information is part of the child class.

I also had an issue with structuring my GitHub repository. The mission should be a single repository with all the tutorial work and consistent folder names.

There are also some inconsistencies in the names in the reviewed repository and a lack of an explicit Tutorial 4 folder. These are things that I need to work on before submitting the final.

The other difficulties were getting from simple Java exercises to GUI programming. More planning is needed for a GUI app, as the program is supposed to react to the user's actions and manage various elements.

4. How the Challenges were overcome

This involved breaking down every problem into smaller steps in order to overcome the programming challenges.

First of all I have to know what classes are needed to solve the problem. I then came up with the attributes and methods that each class would require. Then I developed constructors and objects and tested them out in the main program.

This was a step-by-step approach which made the programs easier to understand and find errors faster.

To inherit, I chose common features of related classes. In my case with people, I thought about what could be common to a Person class, and what can be different with a Student or Lecturer class.

Also I did a few examples more than reading the theory. I found the concepts more clear when I had to access them in practice using a program.

For repository organization, I chose to work in a single repository and to organize it in folders which are tutorials. I learnt that the project should be structured in such a way that another person can easily know where each part of the project is.

Prior to final submission I will also confirm that all required tutorial folders are present, and will fix any inconsistencies in folder names, adding any missing work.

5. Improvements in My Java Programming Skills

Throughout the tutorials, I have definitely gotten better at programming in Java.

Initially I worked on the basics of java syntax and programs. The tutorials helped me get more at ease with creating classes, creating objects, constructors, and methods.

I also enhanced reading and understanding of java source code. I am now able to recognise the function of various classes and understand the creation and use of objects.

The inheritance exercises assisted me in comprehending the relationship among classes. For GUI programming I also had a chance to watch QuizBattleGUI.

Another thing that's improved is my problem solving. Rather than attempting to complete a program all at once, I have come to understand that I should break a problem up into smaller pieces and solve each piece individually.

I also got to learn a little about GitHub, which helped my skills after making my own versions of programs and documenting my projects.

6. Understanding of Object-Oriented Programming Concepts

The tutorials helped me get an understanding of main concepts of Object Oriented Programming.

Classes

A class defines the specification of objects. It specifies the information and actions that things of that kind have.

Objects

An object is a particular instance of a class. Objects enable the program to manipulate particular instances of the information given by a class.

Attributes

The attributes are storing data about the object. For instance, a Student object can have details such as a student's name.

Methods

Methods are actions or behaviour that an object can do. They enable the program to access and manipulate the information of the object.

Constructors

Objects are constructed using constructors, and these can be used to initialise an object's initial values.

Inheritance

An inheritance relationship lets a specialized class inherit all of the attributes of a more general class. Helpful examples that helped me understand this concept were those with Person, Student, Lecturer, employee and appliance related classes.

Abstraction and Class Relationships.

The exercises involving the appliances gave me the idea of how a general concept can be subdivided and how to distinguish between the specialized applications. This is helpful when making larger programs.

In general, the tutorials have shifted my perception of Java from programming instructions to programming programs that are built using objects with data and behavior.

7. Future Learning Plans

My future learning plan will involve further developing my java skills and understanding of Object Oriented Programming.

I want to know more about the concepts of the OOP like Encapsulation, Polymorphism, Interfaces, Inheritance, Abstraction. I wish to gain confidence in designing classes and how they can be used together in a full application.

I am also going to learn more about exception handling, Collection, File Handling and Database connectivity. These topics will assist me in writing bigger and usable java programs.

The other one I would like to work on is Java GUI programming. The GUI quiz application gave me an introduction to interactive programming and I would like to make more advanced programs in the future.

I also want to make myself to be a better programmer at problem solving. I want to break the program down into its component parts so that I can better analyze the problem, design the solution, identify the classes I need, and then implement the program in steps.

I will also enhance my GitHub and Software Development skills. I will adopt consistent folder structures, meaningful file names, clear documentation and helpful commit messages.

I want to develop a complete Java project where I have to incorporate a few OOP concepts in a real-life product. This will enable me to practice something I have learned instead of practicing individual things.

8. Conclusion

The Object Oriented Programming (OOP) tutorials for BIT1123 as a whole have given me a better grasp of Java and Object Oriented Programming (OOP).

In the tutorials I learnt how to create classes and objects, how to use constructors, how to use methods, how to relate to classes and how to use concepts related to inheritance. I also had a bit of experience with GUI programming with the quiz.

Initially, the tutorials were difficult to follow, as I had to shift my thinking from programming statements to objects, classes and relationships between parts of a program. I practiced the exercises and solved problems step by step so that I can get more confident to comprehend and write java programs.

Another lesson I learned was that programming is not just about coding. Good project organization, documentations, management of the GitHub repository and reflection are part of software development. I found it useful to organize my teaching practice in a GitHub repository, as I learned about the need to have a well-defined and structured project.

I have some things that I want to work on in the future such as advanced OOP techniques, exception handling, collection, GUI development, and bigger Java projects. The lessons learned from these tutorials, however, will give me a solid base for my future lessons and programming.

To sum up, this task has helped me to learn a bit more about Java and also about the coding convention used in the profession. I will keep learning and practicing Java and OOP concepts and learn to create more comprehensive and functional software projects with the knowledge acquired from this course.

9. GitHub Repository URL

Repository:
https://github.com/aymanhussein5677/aymanhussein202505010362